

## 2.4 Integrating by Partial Fractions

### Motivating Examples:

Suppose we wish to find  $\int \frac{5x^2 - x - 1}{2x^3 - x^2 - x} dx$  . How could one do this?

Now, suppose we wish to find:  $\int \frac{x^3 + x}{x - 1} dx$  . How could one do this?

**Outline:** In this section our general outline for integrating will be to:

1.

2.

**Case 1 (The denominator is the product of distinct linear factors):**

**Example 1:** Find the exact value of:  $\int_0^1 \frac{x-4}{x^2-5x+6} dx$  .

**Example 2 (you try):** Find  $\int \frac{x^2 + 2x - 1}{x^3 - x} dx$  .

**Case 2 (the denominator is a product of linear factors, some of which are repeated)**

We will illustrate this with an example.

**Example 3:** Find  $\int \frac{x^2 - 5x + 16}{(2x + 1)(x - 2)^2} dx$  .

**Case 3 (the denominator contains irreducible quadratic factors none of which is repeated):**

We illustrate this with an example.

**Example 4:** Find  $\int \frac{4x^2 + 2}{x^3 + x} dx$  .

**Example 5 (you try):**

(a) Find  $\int \frac{2x^2 - x + 4}{x^3 + 4x} dx$  . *Hint: Recall that  $\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right) + C$  where  $a \neq 0$  .*

(b) Find  $\int \frac{x-1}{4x^2 - 4x + 3} dx$  *Hint: The quadratic on the bottom is irreducible. So, you will not be splitting this rational function up. Try to complete the square on the bottom of this fraction.*

**Case 4 (the denominator contains a repeated irreducible quadratic factor)**

**Example 6:** Find a partial fraction decomposition of:  $\frac{x^2 + x + 1}{(x^2 + 1)^2}$ .